DATA MINING & BUSINESS INTELLIGENCE

Laboratory Experiment Record

EXPERIMENT NO. 1

Design of Star Schema and Snowflake Schema for an Airline Reservation System (Flight Ticket Booking)

Student Name	_____
Roll No. / Enrollment No.	_____
Class / Semester	_____
Date of Performance	_____
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1. Aim

To study the concepts of dimensional modelling used in Data Warehousing and Business Intelligence, and to design a Star Schema and a Snowflake Schema for a chosen real-world business domain — the Flight Ticket Booking subsystem of an Airline Reservation System — by identifying the fact table, its measures, the associated dimension tables, and their attributes.

2. Theory

2.1 Data Warehouse and Dimensional Modelling

A Data Warehouse is a subject-oriented, integrated, time-variant and non-volatile collection of data used to support management decision-making. Unlike transactional (OLTP) databases, which are normalized to reduce redundancy and support fast day-to-day operations, a data warehouse is designed for analytical processing (OLAP) — where the priority is fast, intuitive querying and aggregation over large historical datasets.

Dimensional modelling is the design technique used to structure data warehouses so that they are easy to understand and efficient to query. It organizes data into two categories of tables: Fact Tables, which store quantitative, measurable business events, and Dimension Tables, which store descriptive, textual context around those events. Together these form a multidimensional model, commonly implemented as a Star Schema or a Snowflake Schema.

2.2 Fact Table

A fact table is the central table in a dimensional model. Each row represents a single measurable business event (the 'grain' of the fact table), such as one flight ticket booked by one passenger. A fact table typically contains:

- Foreign keys that reference the primary keys of each related dimension table.
- Numeric measures (facts) that can be aggregated — summed, averaged, counted — such as fare amount, number of passengers, or baggage charges.
- Occasionally, degenerate dimensions such as a ticket/PNR number that has no further descriptive attributes of its own.

2.3 Dimension Table

A dimension table stores descriptive attributes that give context and meaning to the facts. Dimension tables are typically wide (many columns), textual, and relatively small compared to the fact table. Examples in an airline reservation system include the Passenger, Flight, Route, and Time dimensions. Dimension attributes are used for filtering, grouping, and labelling reports (e.g., 'total ticket revenue by Route, by Month').

2.4 Star Schema

In a Star Schema, the fact table sits at the centre and is directly connected to a set of fully denormalized dimension tables, so that the diagram resembles a star with the fact table as the centre and dimension tables as the points. Each dimension table contains all of its descriptive attributes in a single flat table, even if some attributes are logically hierarchical (e.g., City and Country both stored inside one Passenger dimension rather than split into separate tables).

Characteristics of a Star Schema:

- Simple structure with a small number of joins between fact and dimension tables.
- Dimension tables are denormalized — they may contain redundant data.
- Faster query performance because fewer joins are required.
- Easier for end users and BI tools to understand and navigate.
- Consumes more storage due to data redundancy in dimensions.

2.5 Snowflake Schema

A Snowflake Schema is an extension of the star schema in which one or more dimension tables are normalized by splitting them into additional related sub-dimension tables, removing redundant data. The resulting diagram looks like a snowflake, with dimension tables branching out into further related tables. For example, the Passenger dimension may be normalized into Passenger, City, and Country tables, and the Flight dimension into Flight, Airline, and Aircraft tables.

Characteristics of a Snowflake Schema:

- Dimension tables are normalized into multiple related tables (typically up to 3NF).
- Reduces data redundancy and improves data integrity.
- Requires more joins, which can reduce query performance compared to a star schema.
- Saves storage space, useful for very large or slowly changing dimensions.
- More complex structure, harder for business users to query directly without BI tool assistance.

2.6 Star Schema vs Snowflake Schema

Parameter	Star Schema	Snowflake Schema
Normalization	Denormalized dimensions	Normalized dimensions (up to 3NF)
Structure	Simple, star-shaped	Complex, snowflake-shaped
Joins required	Fewer joins	More joins
Query performance	Generally faster	Comparatively slower
Storage space	Higher (redundant data)	Lower (less redundancy)
Ease of understanding	Easier for end users	Relatively complex
Data integrity	Lower	Higher
Best suited for	Small/medium warehouses, fast reporting	Large warehouses with big, hierarchical dimensions

3. Domain Description

3.1 Chosen Domain: Airline Reservation System — Flight Ticket Booking

The selected domain for this experiment is the Flight Ticket Booking subsystem of an Airline Reservation System (ARS). Every day, airlines and travel portals process thousands of ticket bookings — passengers select a flight and route, choose a fare class/cabin, book through a particular channel (website, mobile app, travel agent, or airport counter), and are charged a fare made up of a base amount, taxes, seat charges, and baggage charges, sometimes reduced by a discount or promotional offer.

This domain is well suited to dimensional modelling because it naturally produces a very large volume of repetitive, measurable transactions (one row per ticket booked) tied to several descriptive contexts (who the passenger is, which flight and route were booked, which fare class was chosen, which channel was used, and when the booking and travel occurred). Airline revenue management is analytically rich — analysts regularly need answers to questions such as:

- What is the total ticket revenue generated per route per month?
- Which fare class/cabin generates the highest revenue, and what is the average fare per class?
- Which booking channel (online, mobile app, agent, counter) contributes the most bookings and revenue?
- What is the average baggage/seat add-on revenue per aircraft type?
- Which passenger nationality/city generates the most bookings on a given route?

These are precisely the kind of multidimensional, aggregate questions that a Star/Snowflake schema-based data warehouse is designed to answer efficiently, making this an ideal domain for the experiment.

3.2 Business Process / Grain of the Fact Table

The business process being modelled is 'a single ticket booked by a passenger for a single flight.' The grain (the level of detail) of the fact table is therefore defined as: one row per passenger, per flight booking, per ticket issued. Each row captures the numeric outcomes of that single booking event.

4. Fact Table Design — FACT_TICKET_BOOKING

The central fact table stores the measurable, numeric details of every flight ticket booking, along with foreign keys linking it to all relevant dimension tables.

Table: FACT_TICKET_BOOKING

Attribute	Data Type	Description
Ticket_ID (PK)	Integer	Unique surrogate key identifying each ticket/booking transaction (degenerate dimension, business PNR).
Passenger_Key (FK)	Integer	Foreign key referencing DIM_PASSENGER.
Flight_Key (FK)	Integer	Foreign key referencing DIM_FLIGHT.
Route_Key (FK)	Integer	Foreign key referencing DIM_ROUTE.
Aircraft_Key (FK)	Integer	Foreign key referencing DIM_AIRCRAFT.

Attribute	Data Type	Description
Fare_Class_Key (FK)	Integer	Foreign key referencing DIM_FARE_CLASS.
Channel_Key (FK)	Integer	Foreign key referencing DIM_BOOKING_CHANNEL.
Booking_Date_Key (FK)	Integer	Foreign key referencing DIM_TIME (date the ticket was booked).
Travel_Date_Key (FK)	Integer	Foreign key referencing DIM_TIME (date of actual travel/departure).
Number_of_Passengers	Integer (Measure)	Number of passengers covered under this booking record.
Base_Fare	Decimal (Measure)	Base ticket fare before taxes and add-ons.
Tax_Amount	Decimal (Measure)	Total government/airport taxes charged on the ticket.
Seat_Charges	Decimal (Measure)	Additional charges for seat selection/upgrade.
Baggage_Charges	Decimal (Measure)	Additional charges for extra/checked baggage.
Discount_Amount	Decimal (Measure)	Discount or promotional offer amount applied to the booking.
Total_Fare_Amount	Decimal (Measure)	Total amount billed for the ticket (sum of all fare components).
Net_Amount_Paid	Decimal (Measure)	Final amount actually paid by the passenger after discount.

5. Star Schema Design

In the Star Schema version, every dimension table below is kept fully denormalized: all descriptive attributes — including hierarchical ones such as city/country, or airline/aircraft manufacturer — are stored flat within a single dimension table, directly connected to the fact table.

5.1 DIM_PASSENGER

Attribute	Data Type	Description
Passenger_Key (PK)	Integer	Surrogate key for the passenger dimension.
Passenger_ID	Varchar	Natural/business key — passenger's frequent-flyer or booking reference ID.
Passenger_Name	Varchar	Full name of the passenger.
Gender	Varchar	Gender of the passenger.
Date_of_Birth	Date	Passenger's date of birth.
Age	Integer	Passenger's age at the time of booking.
Nationality	Varchar	Passenger's nationality.
Contact_Number	Varchar	Passenger's contact phone number.
Email_ID	Varchar	Passenger's email address.

Attribute	Data Type	Description
City	Varchar	City in which the passenger resides (denormalized).
Country	Varchar	Country in which the passenger resides (denormalized).
Frequent_Flyer_Tier	Varchar	Loyalty programme tier (Silver, Gold, Platinum, etc.).

5.2 DIM_FLIGHT

Attribute	Data Type	Description
Flight_Key (PK)	Integer	Surrogate key for the flight dimension.
Flight_Number	Varchar	Natural key — the flight's unique flight number.
Airline_Name	Varchar	Name of the operating airline (denormalized).
Airline_Code	Varchar	IATA/ICAO code of the airline (denormalized).
Departure_Time	Time	Scheduled departure time of the flight.
Arrival_Time	Time	Scheduled arrival time of the flight.
Flight_Duration_Minutes	Integer	Scheduled duration of the flight in minutes.
Flight_Type	Varchar	Direct or connecting flight indicator.

5.3 DIM_ROUTE

Attribute	Data Type	Description
Route_Key (PK)	Integer	Surrogate key for the route dimension.
Origin_Airport_Code	Varchar	IATA code of the departure airport.
Origin_Airport_Name	Varchar	Name of the departure airport (denormalized).
Origin_City	Varchar	City of the departure airport (denormalized).
Destination_Airport_Code	Varchar	IATA code of the arrival airport.
Destination_Airport_Name	Varchar	Name of the arrival airport (denormalized).
Destination_City	Varchar	City of the arrival airport (denormalized).
Distance_KM	Integer	Flight distance in kilometres for this route.

5.4 DIM_AIRCRAFT

Attribute	Data Type	Description
Aircraft_Key (PK)	Integer	Surrogate key for the aircraft dimension.
Aircraft_Registration_No	Varchar	Unique registration number of the aircraft.
Aircraft_Type	Varchar	Aircraft model/type (e.g., Airbus A320, Boeing 737).
Manufacturer_Name	Varchar	Aircraft manufacturer (denormalized, e.g., Airbus, Boeing).

Attribute	Data Type	Description
Seating_Capacity	Integer	Total number of seats on the aircraft.
Manufacture_Year	Integer	Year the aircraft was manufactured.

5.5 DIM_FARE_CLASS

Attribute	Data Type	Description
Fare_Class_Key (PK)	Integer	Surrogate key for the fare class dimension.
Fare_Class_Code	Varchar	Booking class code (e.g., Y, J, F).
Fare_Class_Name	Varchar	Descriptive fare class name (Economy, Premium Economy, Business, First).
Cabin_Type	Varchar	Cabin category associated with the fare class (denormalized).
Refundable_Flag	Boolean	Indicates whether the fare class is refundable.

5.6 DIM_BOOKING_CHANNEL

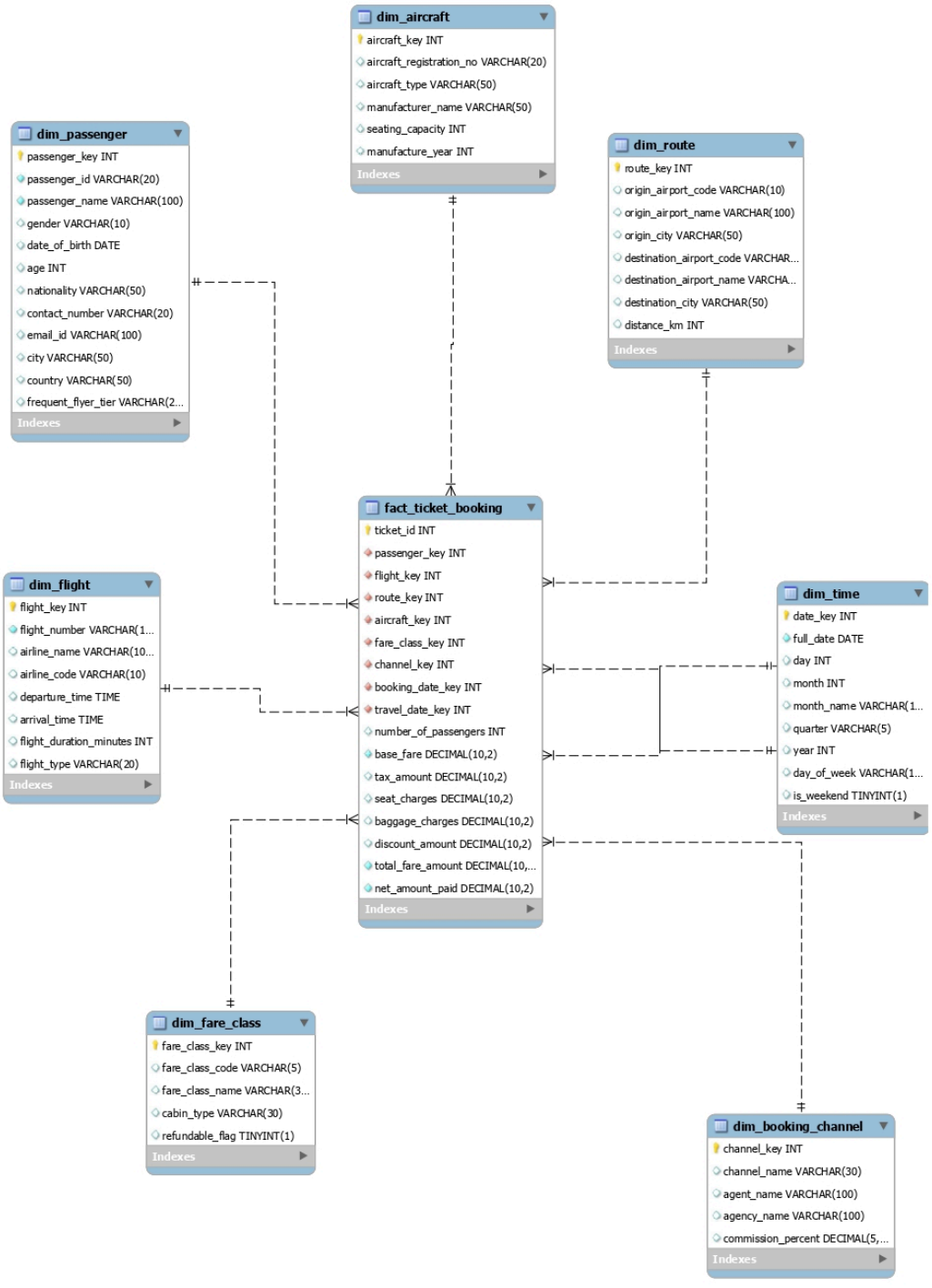
Attribute	Data Type	Description
Channel_Key (PK)	Integer	Surrogate key for the booking channel dimension.
Channel_Name	Varchar	Booking channel used (Website, Mobile App, Travel Agent, Airport Counter).
Agent_Name	Varchar	Name of the travel agent, if applicable (denormalized).
Agency_Name	Varchar	Name of the travel agency, if applicable (denormalized).
Commission_Percent	Decimal	Commission percentage applicable for agent-based bookings.

5.7 DIM_TIME

Attribute	Data Type	Description
Date_Key (PK)	Integer	Surrogate key for the time dimension, one row per calendar date.
Full_Date	Date	Actual calendar date.
Day	Integer	Day of the month.
Month	Integer	Month number (1-12).
Month_Name	Varchar	Name of the month (denormalized, e.g., January).
Quarter	Varchar	Financial/calendar quarter (Q1-Q4).
Year	Integer	Calendar year.
Day_of_Week	Varchar	Name of the weekday.
Is_Weekend	Boolean	Flag indicating whether the date is a weekend.

5.8 Star Schema Diagram

The diagram below should show FACT_TICKET_BOOKING at the centre, with straight single-line connectors radiating outward to each of the seven denormalized dimension tables listed above (DIM_PASSENGER, DIM_FLIGHT, DIM_ROUTE, DIM_AIRCRAFT, DIM_FARE_CLASS, DIM_BOOKING_CHANNEL, DIM_TIME), forming a star shape.



6. Snowflake Schema Design

In the Snowflake Schema version, the Passenger, Flight, Route, Aircraft, and Booking Channel dimensions are normalized further by removing repeating/hierarchical attribute groups into their own related tables. DIM_FARE_CLASS and DIM_TIME remain single tables here since their attributes do not warrant further splitting for this experiment. The fact table and its measures remain unchanged from Section 4.

6.1 Normalized Passenger Branch

DIM_PASSENGER (normalized)

Attribute	Data Type	Description
Passenger_Key (PK)	Integer	Surrogate key for the passenger dimension.
Passenger_ID	Varchar	Natural key — passenger's frequent-flyer or booking reference ID.
Passenger_Name	Varchar	Full name of the passenger.
Gender	Varchar	Gender of the passenger.
Date_of_Birth	Date	Passenger's date of birth.
Contact_Number	Varchar	Passenger's contact phone number.
Email_ID	Varchar	Passenger's email address.
City_Key (FK)	Integer	Foreign key referencing DIM_CITY.
Frequent_Flyer_Tier	Varchar	Loyalty programme tier (Silver, Gold, Platinum, etc.).

DIM_CITY

Attribute	Data Type	Description
City_Key (PK)	Integer	Surrogate key for the city sub-dimension.
City_Name	Varchar	Name of the city.
Country_Key (FK)	Integer	Foreign key referencing DIM_COUNTRY.

DIM_COUNTRY

Attribute	Data Type	Description
Country_Key (PK)	Integer	Surrogate key for the country sub-dimension.
Country_Name	Varchar	Name of the country.
Continent	Varchar	Continent the country belongs to.

6.2 Normalized Flight / Airline Branch

DIM_FLIGHT (normalized)

Attribute	Data Type	Description
Flight_Key (PK)	Integer	Surrogate key for the flight dimension.

Attribute	Data Type	Description
Flight_Number	Varchar	Natural key — the flight's unique flight number.
Departure_Time	Time	Scheduled departure time of the flight.
Arrival_Time	Time	Scheduled arrival time of the flight.
Flight_Duration_Minutes	Integer	Scheduled duration of the flight in minutes.
Flight_Type	Varchar	Direct or connecting flight indicator.
Airline_Key (FK)	Integer	Foreign key referencing DIM_AIRLINE.

DIM_AIRLINE

Attribute	Data Type	Description
Airline_Key (PK)	Integer	Surrogate key for the airline sub-dimension.
Airline_Name	Varchar	Name of the operating airline.
Airline_Code	Varchar	IATA/ICAO code of the airline.
Alliance_Name	Varchar	Airline alliance the carrier belongs to (Star Alliance, OneWorld, etc.).

6.3 Normalized Route / Airport Branch

DIM_ROUTE (normalized)

Attribute	Data Type	Description
Route_Key (PK)	Integer	Surrogate key for the route dimension.
Distance_KM	Integer	Flight distance in kilometres for this route.
Origin_Airport_Key (FK)	Integer	Foreign key referencing DIM_AIRPORT (origin).
Destination_Airport_Key (FK)	Integer	Foreign key referencing DIM_AIRPORT (destination).

DIM_AIRPORT

Attribute	Data Type	Description
Airport_Key (PK)	Integer	Surrogate key for the airport sub-dimension (referenced twice by DIM_ROUTE — once for origin, once for destination).
Airport_Code	Varchar	IATA code of the airport.
Airport_Name	Varchar	Full name of the airport.
City_Key (FK)	Integer	Foreign key referencing DIM_CITY (shared with the passenger branch).

6.4 Normalized Aircraft Branch

DIM_AIRCRAFT (normalized)

Attribute	Data Type	Description
Aircraft_Key (PK)	Integer	Surrogate key for the aircraft dimension.
Aircraft_Registration_No	Varchar	Unique registration number of the aircraft.
Aircraft_Type	Varchar	Aircraft model/type (e.g., Airbus A320, Boeing 737).
Seating_Capacity	Integer	Total number of seats on the aircraft.
Manufacture_Year	Integer	Year the aircraft was manufactured.
Manufacturer_Key (FK)	Integer	Foreign key referencing DIM_MANUFACTURER.

DIM_MANUFACTURER

Attribute	Data Type	Description
Manufacturer_Key (PK)	Integer	Surrogate key for the manufacturer sub-dimension.
Manufacturer_Name	Varchar	Aircraft manufacturer name (e.g., Airbus, Boeing).
Headquarters_Country	Varchar	Country where the manufacturer is headquartered.

6.5 Normalized Booking Channel Branch

DIM_BOOKING_CHANNEL (normalized)

Attribute	Data Type	Description
Channel_Key (PK)	Integer	Surrogate key for the booking channel dimension.
Channel_Name	Varchar	Booking channel used (Website, Mobile App, Travel Agent, Airport Counter).
Agent_Key (FK)	Integer	Foreign key referencing DIM_AGENT (nullable for direct/online bookings).

DIM_AGENT

Attribute	Data Type	Description
Agent_Key (PK)	Integer	Surrogate key for the agent sub-dimension.
Agent_Name	Varchar	Name of the individual travel agent.
Commission_Percent	Decimal	Commission percentage applicable for this agent's bookings.
Agency_Key (FK)	Integer	Foreign key referencing DIM_AGENCY.

DIM_AGENCY

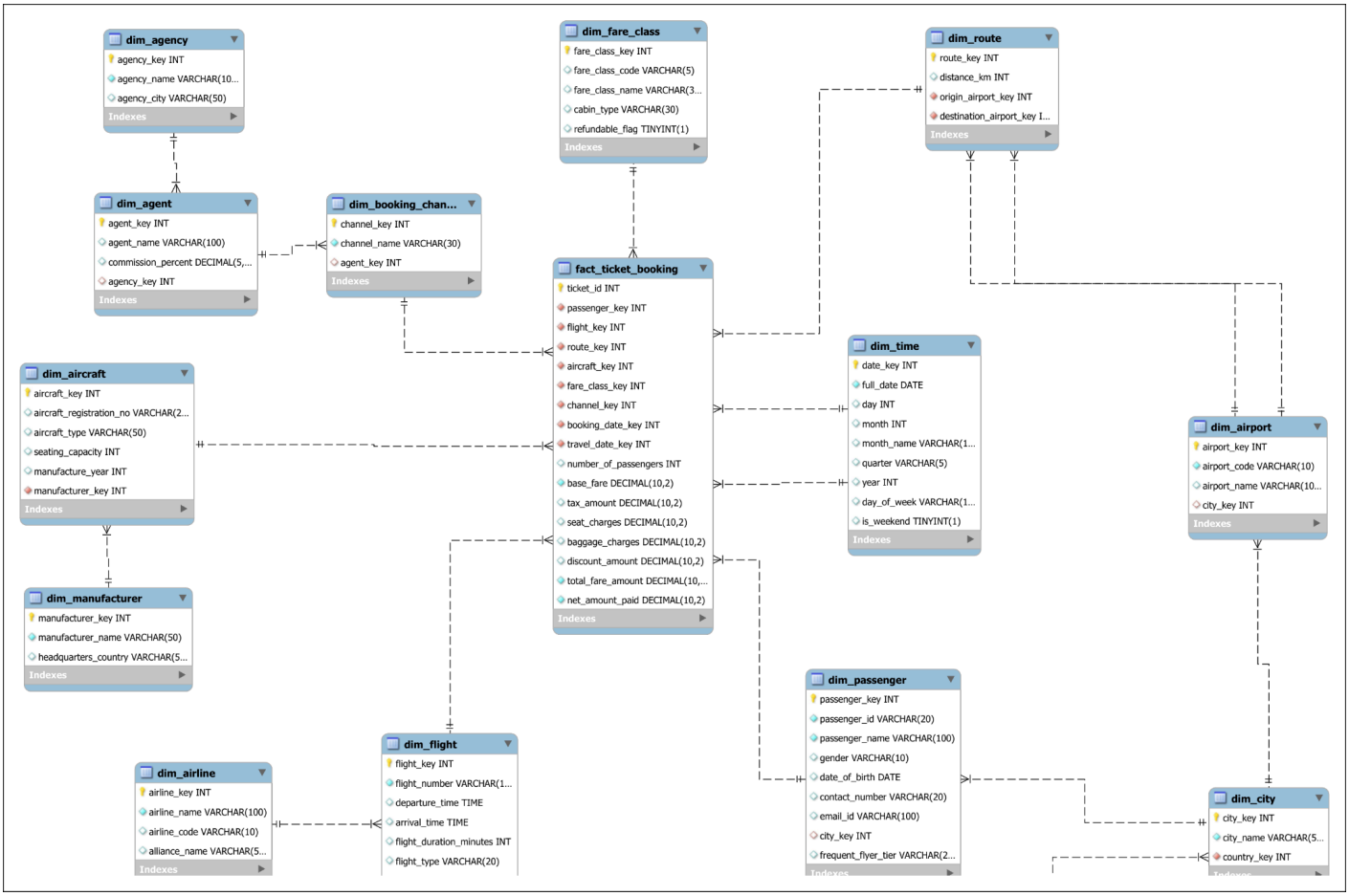
Attribute	Data Type	Description
Agency_Key (PK)	Integer	Surrogate key for the agency sub-dimension.
Agency_Name	Varchar	Name of the travel agency.
Agency_City	Varchar	City in which the agency operates.

6.6 Unchanged Dimensions

DIM_FARE_CLASS and DIM_TIME retain the same structure described in Sections 5.5 and 5.7, as their attributes are already atomic and do not represent a hierarchy worth normalizing for this system.

6.7 Snowflake Schema Diagram

The diagram below should show FACT_TICKET_BOOKING at the centre connected to the primary dimension tables (DIM_PASSENGER, DIM_FLIGHT, DIM_ROUTE, DIM_AIRCRAFT, DIM_FARE_CLASS, DIM_BOOKING_CHANNEL, DIM_TIME) exactly as in the star schema, but each of these primary dimensions should then branch outward one level further to their normalized sub-dimension tables (e.g., DIM_PASSENGER → DIM_CITY → DIM_COUNTRY; DIM_FLIGHT → DIM_AIRLINE; DIM_ROUTE → DIM_AIRPORT (origin and destination) → DIM_CITY; DIM_AIRCRAFT → DIM_MANUFACTURER; DIM_BOOKING_CHANNEL → DIM_AGENT → DIM_AGENCY), giving the overall diagram its characteristic snowflake shape.



Conclusion

This experiment applied dimensional modelling to the Airline Reservation System using a fact table and seven dimension tables. The **Star Schema** provides simple, fast reporting, while the **Snowflake Schema** reduces redundancy and improves data integrity. Overall, Star Schema is better for routine reporting, whereas Snowflake Schema suits large and complex data warehouses.